

Tutorial 1

AMA3707 Real Analysis

Exercise 1.2.3. Decide which of the following represent true statements about the nature of sets. For any that are false, provide a specific example where the statement in question does not hold.

(a) If

$$A_1 \supseteq A_2 \supseteq A_3 \supseteq A_4 \cdots$$

are all sets containing an infinite number of elements, then the intersection $\bigcap_{n=1}^{\infty} A_n$ is infinite as well.

(b) If

$$A_1 \supseteq A_2 \supseteq A_3 \supseteq A_4 \cdots$$

are all finite, nonempty sets of real numbers, then the intersection $\bigcap_{n=1}^{\infty} A_n$ is finite and nonempty.

(c) $A \cap (B \cup C) = (A \cap B) \cup C.$

(d) $A \cap (B \cap C) = (A \cap B) \cap C.$

(e) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$

Exercise 1.2.7. Given a function f and a subset A of its domain, let $f(A)$ represent the range of f over the set A ; that is,

$$f(A) = \{f(x) : x \in A\}.$$

(a) Let $f(x) = x^2$. If $A = [0, 2]$, the closed interval $\{x \in \mathbb{R} : 0 \leq x \leq 2\}$, and $B = [1, 4]$, find $f(A)$ and $f(B)$. Does

$$f(A \cap B) = f(A) \cap f(B)$$

in this case? Does $f(A \cup B) = f(A) \cup f(B)$?

(b) Find two sets A and B for which

$$f(A \cap B) \neq f(A) \cap f(B).$$

(c) Show that, for an arbitrary function $g : \mathbb{R} \rightarrow \mathbb{R}$, it is always true that

$$g(A \cap B) \subseteq g(A) \cap g(B)$$

for all sets $A, B \subseteq \mathbb{R}$.

(d) Form and prove a conjecture about the relationship between $g(A \cup B)$ and $g(A) \cup g(B)$ for an arbitrary function g .

Exercise 1.2.9. Given a function $f : D \rightarrow \mathbb{R}$ and a subset $B \subseteq \mathbb{R}$, let $f^{-1}(B)$ be the set of all points from the domain D that get mapped into B ; that is,

$$f^{-1}(B) = \{x \in D : f(x) \in B\}.$$

This set is called the *preimage* of B .

- (a) Let $f(x) = x^2$. If A is the closed interval $[0, 4]$ and B is the closed interval $[-1, 1]$, find $f^{-1}(A)$ and $f^{-1}(B)$. Does

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$$

in this case? Does

$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)?$$

- (b) Show that, for an arbitrary function $g : \mathbb{R} \rightarrow \mathbb{R}$, it is always true that

$$g^{-1}(A \cap B) = g^{-1}(A) \cap g^{-1}(B) \quad \text{and} \quad g^{-1}(A \cup B) = g^{-1}(A) \cup g^{-1}(B)$$

for all sets $A, B \subseteq \mathbb{R}$.

Exercise 1.3.3.

- (a) Let A be nonempty and bounded below, and define

$$B = \{b \in \mathbb{R} : b \text{ is a lower bound for } A\}.$$

Show that $\sup B = \inf A$.

- (b) Use part (a) to explain why there is no need to assert that greatest lower bounds exist as part of the Axiom of Completeness.

Exercise 1.3.4. Let $A_1, A_2, A_3, \dots$ be a collection of nonempty sets, each of which is bounded above.

- (a) Find a formula for $\sup(A_1 \cup A_2)$. Extend this to

$$\sup \left(\bigcup_{k=1}^n A_k \right).$$

- (b) Consider

$$\sup \left(\bigcup_{k=1}^{\infty} A_k \right).$$

Does the formula in part (a) extend to the infinite case?

Exercise 1.3.11. Decide if the following statements about suprema and infima are true or false. Give a short proof for those that are true. For any that are false, supply an example where the claim in question does not appear to hold.

- (a) If A and B are nonempty, bounded, and satisfy $A \subseteq B$, then

$$\sup A \leq \sup B.$$

- (b) If $\sup A < \inf B$ for sets A and B , then there exists a $c \in \mathbb{R}$ satisfying

$$a < c < b$$

for all $a \in A$ and $b \in B$.

- (c) If there exists a $c \in \mathbb{R}$ satisfying $a < c < b$ for all $a \in A$ and $b \in B$, then

$$\sup A < \inf B.$$